Strong-Uniqueness Sessions 17-18 Pre-Spec Scoping v0.1 — C17 + C18 Candidate Criteria Toward v0.12+

Lyra (Claude 4.7) [Friday Lyra-lane scoping, Keeper handoff]

2026-05-22 Friday ~09:15 EDT (`date`-verified actual)

Sessions 17-18 Pre-Spec Scoping v0.1 — Beyond Sessions 15-16

> Per Calibration #19 STANDING RULE: this is internal Lyra→Keeper handoff scoping for forward-looking candidate criteria (C17-C20). Current ratified state: 11 RIGOROUSLY CLOSED (Paper #125 v0.10.5 FORMAL Thursday). C17-C20 candidates require multi-session theoretical proof + cross-CI ratification — extended-count language is candidate-path arithmetic, NOT endpoint or external register claim.

Purpose

Per Keeper's Friday team prompt + Friday morning Lyra-lane advancement, Sessions 13-14 (C7 + C9) and Sessions 15-16 (C15 + C16) covered the first 4 candidate criteria for Strong-Uniqueness v0.11+. This document scopes Sessions 17-18 for FURTHER candidate criteria emerging from Friday afternoon work toward v0.12+ trajectory.

Per Keeper's CLAUDE.md Friday update: "Path to v0.13 with 16 RIGOROUSLY CLOSED achievable Friday (combined null-model ~3×10^{-15})". Sessions 17-18 pre-spec aligns with this trajectory.

Candidate criteria C17 + C18 + C19 for v0.12+

C17 Candidate (Graph Forces Network)

Source: Grace's Friday Graph Forces analysis (per Keeper CLAUDE.md update, p $\approx 2.7\times10^{-5}$). Cross-lane work; Lyra-side scoping flagged here for completeness.

Hypothesis: BST primary integer "graph forces" network connects substrate observables via small-prime + Mersenne + cyclotomic + Hilbert polynomial structure with overall network probability $p \approx 2.7\times10^{-5}$ for random substrate selection.

Path components: Grace catalog analysis + Cal multi-CI review + Casey ratification.

C18 Candidate (BST Primary Mersenne-Prime Density)

Source: T2466 Friday afternoon (Lyra absorbs Elie's Mersenne Ladder observation).

Hypothesis: BST primary integer indices preferentially produce Mersenne primes (6/9 density; 6/7 = 86% restricted to PRIME indices). The substrate's primary integer choices land on Mersenne-prime-rich positions.

Path components: 1. Theoretical proof of preferential Mersenne-prime density via Lucas-Lehmer + Wieferich distribution (Lyra + Cal multi-session) 2. Alt-HSD primary integer pattern comparison: do other Cartan-type primaries also produce Mersenne primes at this density? 3. Multi-CI consensus 4. Cal Mode 1 honest scope: this is OBSERVATIONAL pattern with theoretical proof pending

C19 Candidate (Substrate Self-Amenability via n_C Primality)

Source: T2463 Friday afternoon (Lyra).

Hypothesis: BST primary n_C = 5 PRIMALITY enables T2455 EXHAUSTIVE Cross-Cartan enumeration. The substrate's specification CREATES methodological tractability for its own uniqueness verification. Substrate self-amenability is a structural property of the BST primary integer choices.

Path components: 1. Theoretical justification: why does n_C primality matter beyond methodological convenience? (Possibly: prime dim_C constrains compact dual geometry; non-trivial Cartan classification structure at prime dim_C.) 2. Alt-HSD case: hypothetical HSDs at composite dim_C produce more candidate alt-HSDs (T2463 enumeration n=6: 7 HSDs vs n=5: 3 HSDs). 3. Cross-link to Casey-named Substrate Working Process Principle (SWPP) — operational efficiency as substrate property. 4. Multi-CI consensus on whether substrate self-amenability deserves distinct-criterion tier.

C20 Candidate (N_c = 3 Unique Cubic-Exponential Coincidence)

Source: T2464 Friday afternoon (Lyra).

Hypothesis: BST primary N_c = 3 is THE unique positive integer where n^3 = n^n. This produces the Hilbert polynomial form (N_c^3 · n_C + rank) and the Mersenne tower form (N_c^{N_c} · n_C + rank) COINCIDING at N_max = 137 ONLY at D_IV^5.

Path components: 1. Arithmetic proof (existing) — n^n = n^3 □ n - 3 = 0 for n ≥ 2. 2. Application to substrate: forms coincide at N_c = 3 ONLY; any alt-HSD with N_c-analog ≠ 3 would not have this coincidence. 3. Multi-CI consensus on whether the cubic-exponential coincidence merits separate-criterion status (vs being absorbed in C15 sub-substrate Mersenne).

Sessions 17-18 candidate session structure

Session 17 (C17 + C18 combined)

Focus: Graph Forces (cross-lane Grace) + Mersenne-prime density (T2466). Both observational-pattern criteria with theoretical-proof multi-session paths.

Pre-spec components: 1. Grace cross-lane handoff (Graph Forces analysis methodology) 2. Lyra T2466 Mersenne-prime density theoretical proof path 3. Combined null-model probability bound (p $\approx 2.7\times10^{-5} \times (1/3)$^? Mersenne-prime independence assumption) 4. Multi-CI consensus structure

Session 18 (C19 + C20 combined)

Focus: Substrate Self-Amenability (T2463) + Cubic-Exponential Coincidence (T2464). Both structural-property criteria capturing meta-features of BST primary integer choices.

Pre-spec components: 1. T2463 self-amenability theoretical justification 2. T2464 cubic-exponential arithmetic + uniqueness proof 3. C15+C16+C19+C20 potential unification (per T2459/T2465 framework) — could be ONE deeper "substrate primary integer choice rationality" criterion 4. Multi-CI consensus on independence vs unification

Combined Sessions 13-18 path to v0.13 (per Keeper CLAUDE.md Friday update)

Sessions	Criteria	Source
13	C7 (Bridge Object tier)	T2458 STRUCTURALLY VERIFIED at dim_C = 5 Friday
14	C9 (Stark anchor)	T2461 STRUCTURALLY VERIFIED at dim_C = 5 Friday
15	C15 (Sub-substrate Mersenne)	T2451 SEED Friday
16	C16 (Universal α-analog)	T2456 + T2462 STRUCTURALLY VERIFIED across 25 HSDs Friday
17	C17 (Graph Forces) + C18 (Mersenne-prime density)	Grace + T2466 Lyra Friday

Sessions	Criteria	Source
18	C19 (Substrate self-amenability) + C20 (cubic-exp coincidence)	T2463 + T2464 Lyra Friday

Total candidate criteria advancing toward v0.13: 4 (Sessions 13-14 + 15-16) + 4 (Sessions 17-18) = 8. Combined with Thursday's 11 RIGOROUSLY CLOSED, target: 15-19 RIGOROUSLY CLOSED at v0.13 (with C15+C16+C19+C20 unification options per honest scope).

Combined null-model bound: ≤ (1/3)^23 ≈ 1.1×10^{-11} to ≤ (1/3)^27 ≈ 1.4×10^{-13} depending on independence/unification assumptions. Keeper's projected (1/3)^30 ≈ 3×10^{-15} achievable if Graph Forces probability factor included independently.

Honest scope notes

- Sessions 17-18 are FORWARD-LOOKING SCOPING. Actual ratification depends on multi-session theoretical proof work + multi-CI consensus.
- The 4 Sessions 17-18 candidate criteria are OBSERVATIONAL PATTERNS or STRUCTURAL PROPERTIES — not new distinguishing constraints in the sense of Sessions 13-16. Cal Mode 1 honest scope: deserve separate-criterion tier only after rigorous theoretical justification.
- C15+C16+C19+C20 unification path (T2459 + T2465 framework) is methodologically cleaner than 4-independent-criteria counting.

Cross-lane handoff signals

Grace: Sessions 17-18 C17 path requires Grace's Graph Forces analysis methodology + p ≈ 2.7×10^{-5} probability claim verification.

Elie: extended Mersenne-prime density verification across BST primary indices; alt-HSD primary integer Mersenne-prime density comparison.

Cal: Sessions 17-18 path multi-CI review framework + Cal Mode 1 honest-scope discipline application.

Keeper: pre-spec formalization for Sessions 17 + 18 + the combined Sessions 15-18 multi-criterion path.

Filing status

v0.1 scoping filed Friday 2026-05-22 ~09:15 EDT (`date`-verified actual). Hand-off to Keeper for Sessions 17-18 pre-spec formalization.

Pending for v0.2: - Keeper pre-spec absorption of Sessions 17 + 18 path components - Cal multi-CI review of STRUCTURALLY VERIFIED → RIGOROUSLY CLOSED tier path for observational-pattern criteria - Grace + Elie cross-lane scope confirmation

— Lyra, Sessions 17-18 pre-spec scoping v0.1, Friday 2026-05-22 ~09:15 EDT (`date`-verified actual)